Supplement of AOI-Net: Structural Face AOI-Guided Eye-Gaze Track Representation Learning for Autism Spectrum Disorder Detection

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I. Dataset

This study utilizes eight eye-movement datasets, namely Speaking, Walking1, Walking2, Helicopter, Baby, Tablet, Attention, and Sad, collected by the Shenzhen Maternity and Child Healthcare Hospital. These datasets were designed to capture children's gaze behaviors under different dynamic video viewing scenarios, providing diverse eye-movement patterns for ASD-related analysis. The ASD group was diagnosed through comprehensive clinical evaluations, including standardized behavioral assessments and expert evaluations conducted by licensed clinicians. The TD group was screened to exclude known neurodevelopmental disorders. Detailed demographic and clinical characteristics of the participants are summarized in Table s-I. The observed male predominance in the dataset is consistent with the demographic characteristics commonly reported in clinically diagnosed ASD cohorts.

During data collection, participants were seated at a controlled viewing distance of 57–65 cm, and eye movements were recorded using an SMI eye-tracking system with a sampling rate of 250 Hz. A standard calibration procedure was performed before each recording session to ensure gaze measurement accuracy. All participants were presented with identical dynamic video stimuli under the same experimental protocol, with each video corresponding to a fixed-duration viewing task. The hardware configuration and key preprocessing parameters are summarized in Table s-II.

The raw gaze signals were converted into fixation sequences using a standard fixation detection procedure, where a fixation was defined as consecutive gaze samples lasting longer than 100 ms. During preprocessing, invalid gaze samples caused by blinks or tracking loss were removed, and gaze points outside the valid display region were filtered. The resulting fixation sequences were subsequently mapped to predefined areas of interest (AOIs), which were manually annotated on video frames and consistently applied across all participants. These AOI-based representations preserve the semantic organization of visual regions and serve as the structural input for the proposed framework.

The objective of this dataset is to distinguish children with ASD from their typically developing (TD) peers, providing a supportive tool for ASD screening. As an initial step in the clinical assessment, ASD-versus-TD identification provides a foundation for subsequent evaluations, including subtype characterization, comorbidity analysis, and individualized assessment. The numbers of ASD and TD samples for each experimental paradigm, together with the corresponding imbalance ratios ($\mathrm{IR} = N_{\mathrm{ASD}}/N_{\mathrm{TD}}$), are reported in Table s-III.

TABLE s-I
PARTICIPANT DEMOGRAPHICS AND CLINICAL CHARACTERISTICS OF THE DATASET.

Characteristic	Value
Age of ASD group (years)	1.93 ± 0.73
Age of TD group (years)	1.74 ± 0.57
Gender distribution	Male: 73.9%, Female: 26.1%
Male-to-female ratio	2.84:1

TABLE s-II
EYE-TRACKING HARDWARE CONFIGURATION AND PREPROCESSING PARAMETERS.

Category	Description
Eye-tracking device	SMI eye tracker
Sampling rate	250 Hz
Eye tracking mode	Binocular tracking
Operating distance	57–65 cm
Stimulus type	Dynamic video stimuli
Fixation definition	Consecutive gaze samples exceeding 100 ms
Calibration protocol	Smart calibration procedure provided by the SMI eye-tracking system

TABLE s-III
DATASET COMPOSITION: THE TABLE SUMMARIZES THE NUMBER OF PARTICIPANTS IN THE DATASET, CATEGORIZED BY ASD AND TD INDIVIDUALS. IR DENOTES THE IMBALANCE RATIO.

Index	Dataset	Participants	ASD	TD	IR
1	Speaking	1390	966	424	2.28
2	Walking1	1356	939	417	2.25
3	Walking2	1380	957	423	2.26
4	Helicopter	1375	955	420	2.27
5	Baby	1348	939	409	2.30
6	Tablet	1372	954	418	2.28
7	Attention	1349	939	410	2.29
8	Sad	1277	882	395	2.23

II. Algorithm

Algorithm 1 summaries the overall training procedure. Specifically, in **Temporal Encoding** (Line 6), the temporal encoder processes the fixation sequence to capture fixation transitions and motion dynamics, producing temporal gaze representations. Next, during **AOI Graph Construction** (Lines 7–11), a graph is constructed by integrating temporal edges that preserve sequential fixation transitions and AOI-aware semantic edges that connect fixation points belonging to the same region of interest. This graph jointly models temporal continuity and spatial correlations among fixations. The resulting graph is then processed in **Structural Encoding** (Line 12) by a graph neural network to obtain structured gaze representa-

Algorithm 1: Training Procedure of AOI-Net.

Input: Fixation sequence $S = \{s_1, ..., s_T\}$, AOI labels $A = \{a_1, ..., a_T\}$
Output: Model parameters θ
1 Initialize parameters θ;
2 **for** *each training epoch* **do**
3   **for** *each mini-batch* (S, A, y) **do**
4   **Temporal encoding:** $\mathbf{h} \leftarrow \text{TemporalEncoder}(S)$;
5   **AOI graph construction:**
6   Add temporal edges: $(v_t, v_{t+1}) \in \mathcal{E}_{Temp}$;
7   Add AOI edges: $(v_i, v_j) \in \mathcal{E}_{AOI}$ if $a_i = a_j$;
8   Construct edge set: $\mathcal{E} \leftarrow \mathcal{E}_{Temp} \cup \mathcal{E}_{AOI}$;
9   Form graph structure: $\mathcal{G} \leftarrow (\mathcal{V}, \mathcal{E})$;
10   **Structural encoding:** $\mathbf{g} \leftarrow \text{GNN}(\mathcal{G})$;
11   **Dual-expert fusion:** $(\omega_{TE}, \omega_{SE}) \leftarrow \text{Softmax}(\text{Gate}(S))$;
12     $\mathbf{z} \leftarrow \omega_{TE} \cdot \mathbf{h} + \omega_{SE} \cdot \mathbf{g}$;
13   **Classification:** $\hat{y} \leftarrow \text{Classifier}(\mathbf{z})$;
14   **Loss computation:** $L_{Im} \leftarrow \eta L_{CaR} + (1-\eta) L_{Ia}$;
15     $L_{Total} \leftarrow L_{Im} + \mu L_{LB}$;
16   **Parameter update:** $\theta \leftarrow \text{Adam}(\theta, \nabla L_{Total})$;
17   **end**
18 **end**

tions. Finally, in **Dual-Expert Fusion** (Lines 13–14), temporal and structural representations are adaptively integrated through a gating-based fusion mechanism, where fusion weights are dynamically learned to balance the complementary contributions of the two branches. By jointly leveraging temporal dynamics and AOI-aware structural dependencies, AOI-Net learns discriminative and structured gaze representations for reliable ASD screening.

III. Supplementary Experiments

A. Sensitivity Analysis of the Hyperparameter

To evaluate the stability of AOI-Net in practical applications, we conducted a sensitivity analysis on the hyperparameter η, which governs the trade-off between the Class-aware Representation loss and the Imbalance-aware loss. As illustrated in Fig. s-1, the model's performance in terms of both ACC and F1-score remains remarkably stable across a broad range of η values.

The robustness of the proposed method with respect to the hyperparameter η is evaluated through a sensitivity analysis, in which η is varied while all other hyperparameters are fixed. The results presented in Fig. s-1 show that the performance exhibits limited variation across a wide range of η values on all datasets, demonstrating strong robustness. The stability of the hyperparameter is particularly advantageous for clinical deployment, as it suggests that AOI-Net does not require exhaustive hyperparameter tuning to achieve reliable results on new datasets. In particular, setting η to 0.25 yields the best average performance and is therefore selected for subsequent experiments.

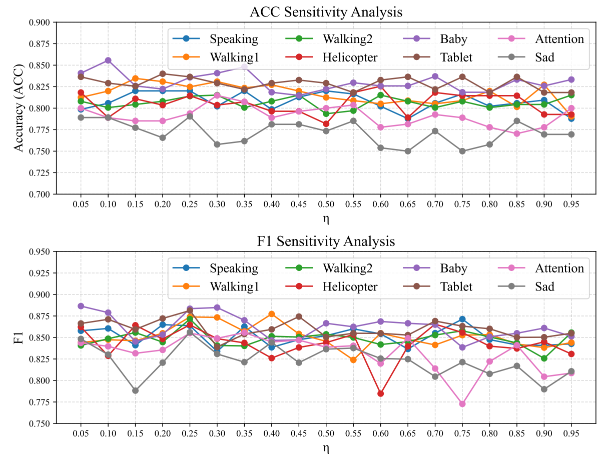


Fig. s-1. Sensitivity analysis of the hyperparameter η in terms of ACC and F1-score.

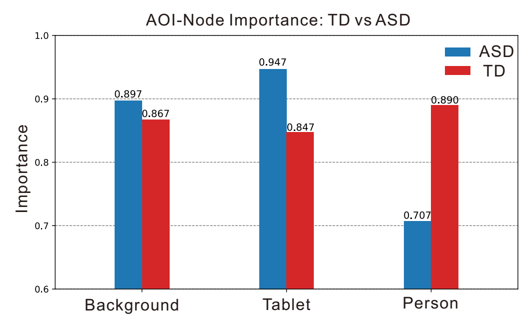


Fig. s-2. Average AOI node importance for the ASD and TD groups computed using a gradient-based interpretation method. The importance score reflects each AOI node's contribution to the final classification output, with higher values indicating greater influence on the model decision.

B. Analysis of AOI Importance and Prototype Behavior

To enhance the clinical interpretability of the proposed framework, we further analyze the contribution of each AOI node using a gradient-based attribution method, which quantifies how strongly the features of each AOI node contribute to the final decision. The result of the Tablet dataset is shown in Fig. s-2.

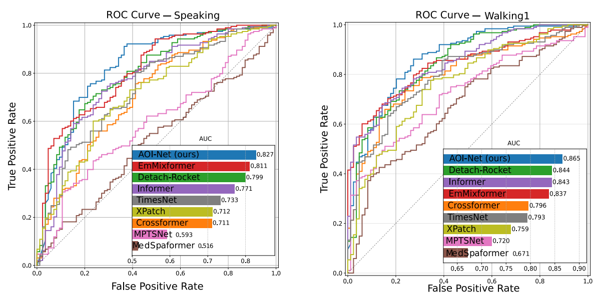
A distinct divergence in importance distribution can be observed between the two groups. For TD samples, the model predominantly assigns higher importance to the AOI corresponding to the "Person", indicating a stronger reliance on socially relevant cues. In contrast, when discriminating ASD samples, the network places greater emphasis on non-social regions, particularly the "Background" and the "Tablet". This tendency is consistent with well-documented atypical attentional preferences in ASD, where individuals are more likely to allocate gaze to objects or environmental details rather than socially salient information.

C. Analysis of Fusion Strategies and Gating Mechanisms

To validate the effectiveness of the proposed gated-sum MoE fusion strategy, we compared it with several representative fusion schemes, including concatenation, mean fusion,

TABLE s-IV
FUSION STRATEGIES AND GATE ACTIVATIONS IN THE MOE MODULE. **BOLD** AND UNDERLINED DENOTE THE BEST AND SECOND-BEST RESULTS.

Dataset	Metric	Fusion Strategy			Activation Function	
		Cross-attention	Concatenation	Mean Fusion	AOI-Net (Sigmoid)	AOI-Net (Softmax)
Speaking	ACC	0.8141	0.8034	**0.8249**	0.8225	0.8201
	F1	0.8698	0.8612	**0.8797**	0.8769	0.8637
Walking1	ACC	0.7953	0.8150	0.8162	**0.8297**	0.8238
	F1	0.8455	0.8646	0.8611	**0.8775**	0.8739
Walking2	ACC	0.8007	0.7971	0.8056	0.8068	**0.8140**
	F1	0.8615	0.8519	0.8639	0.8680	**0.8714**
Helicopter	ACC	0.7830	0.7964	0.8085	0.8097	**0.8145**
	F1	0.8444	0.8542	0.8617	**0.8654**	0.8647
Baby	ACC	0.8235	0.8111	0.8321	0.8185	**0.8358**
	F1	0.8706	0.8652	0.8790	0.8689	**0.8835**
Tablet	ACC	0.8182	0.8206	0.8327	0.8339	**0.8364**
	F1	0.8649	0.8666	0.8746	0.8798	**0.8813**
Attention	ACC	0.7580	0.7741	0.7790	0.7877	**0.7938**
	F1	0.8319	0.8407	0.8494	0.8483	**0.8551**
Sad	ACC	0.7461	0.7578	0.7721	0.7760	**0.7904**
	F1	0.8179	0.8225	0.8439	0.8451	**0.8561**
Average Rank	ACC	4.50	4.38	2.63	2.13	**1.38**
	F1	4.38	4.25	2.63	2.13	**1.63**



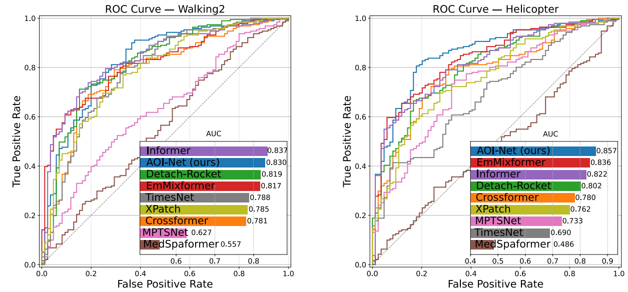
(a) (b)

Fig. s-3. ROC curves and AUC of different methods on the (a) Speaking and (b) Walking1 datasets.

and cross-attention. We also replaced the softmax-based gating mechanism with sigmoid activation to investigate the influence of gate functions. The results are summarized in Table s-IV. The proposed AOI-Net achieves consistent performance improvements with different gating mechanisms, demonstrating the effectiveness of adaptive expert fusion for integrating temporal and AOI representations. Fixed fusion strategies, including concatenation and mean fusion, remain competitive in some cases but cannot adjust expert contributions according to different gaze patterns. Cross-attention-based fusion does not consistently outperform the lightweight gating mechanism, possibly because excessive feature interaction may weaken the specialization of temporal and structural experts. The sigmoid-based variant achieves comparable results, while softmax gating provides more stable improvements across tasks. This indicates that the main benefit comes from adaptive expert weighting rather than a specific activation function.

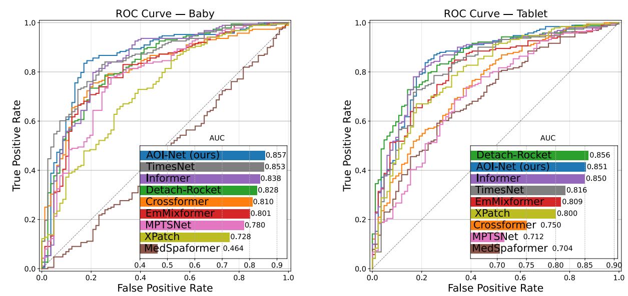
D. ROC Analysis on All Datasets

Due to space limitations, only the representative ROC curves on the Speaking and Walking1 datasets were reported in the main text. To further examine the screening capability of different methods across varying decision thresholds, we provide the ROC curves for all eight datasets in Figs. s-3–s-6. As shown in these figures, the ROC curves of AOI-Net consistently dominate those of competing approaches over most operating regions on every dataset, indicating a superior and stable balance between Sensitivity and Specificity. Moreover, AOI-Net achieves the largest AUC values across all datasets, demonstrating its stronger and more robust discriminative ability between ASD and TD under diverse experimental paradigms.



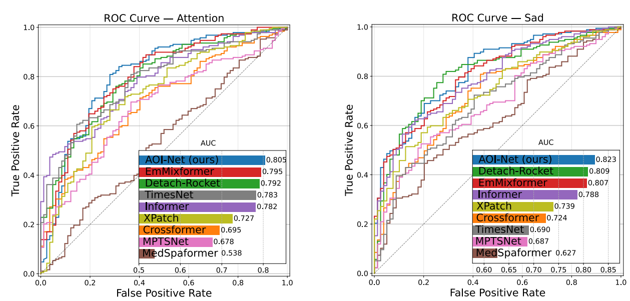
(a) (b)

Fig. s-4. ROC curves and AUC of different methods on the (a) Walking2 and (b) Helicopter datasets.



(a) (b)

Fig. s-5. ROC curves and AUC of different methods on the (a) Baby and (b) Tablet datasets.



(a) (b)

Fig. s-6. ROC curves and AUC of different methods on the (a) Attention and (b) Sad datasets.